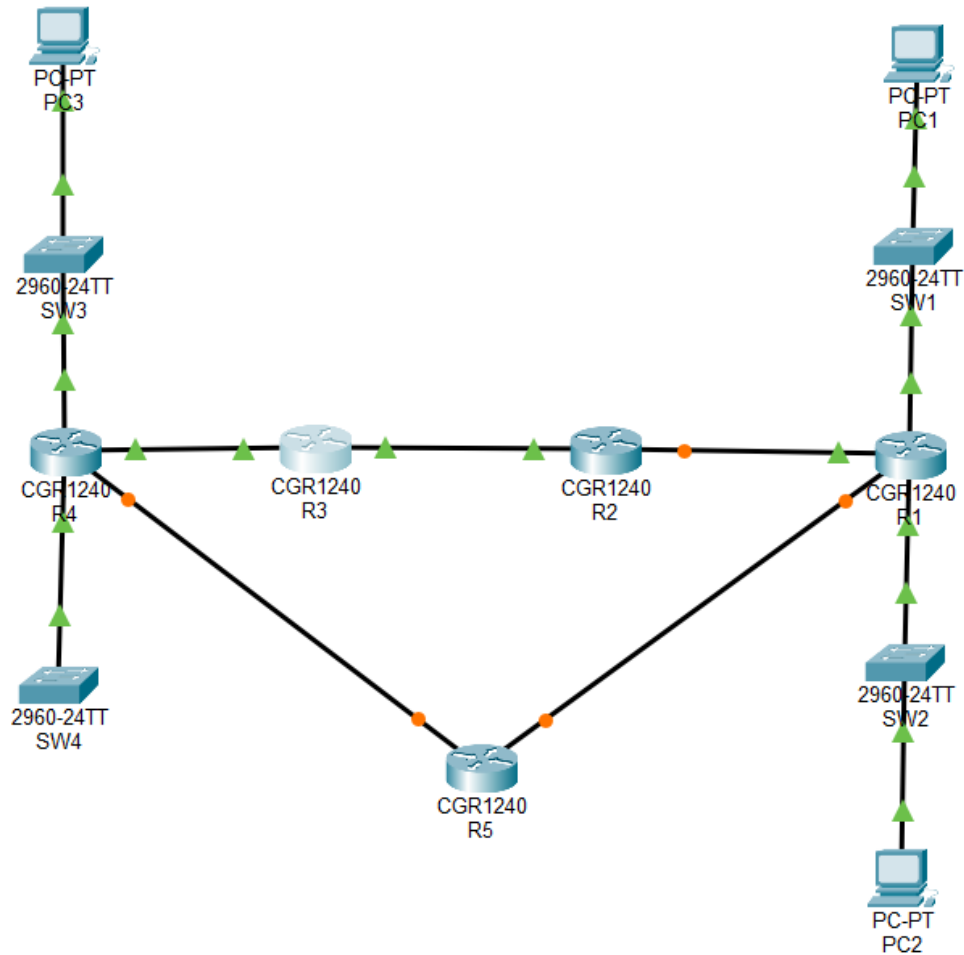


Week 7: Routing Fundamentals (10 Marks)



52. R1:

```
Router>ena
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R1
R1(config)#
R1(config)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console
```

R2:

```
Router>ena
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R2
R2(config)#end
R2#
%SYS-5-CONFIG_I: Configured from console by console
```

R3:

```
Router>ena
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R3
R3(config)#end
R3#
%SYS-5-CONFIG_I: Configured from console by console
```

R4:

```
Router>ena
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R4
R4(config)#end
R4#
%SYS-5-CONFIG_I: Configured from console by console
```

R5:

```
Router>ena
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R5
R5(config)#end
R5#
%SYS-5-CONFIG_I: Configured from console by console
```

53. configuring IP addresses on R1

```
R1>ena
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#interface FastEthernet2/3
R1(config-if)#no switchport
R1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/3, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/3, changed state to up

R1(config-if)#ip address 10.0.0.0 255.255.255.0
Bad mask /24 for address 10.0.0.0
R1(config-if)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#interface FastEthernet2/3
R1(config-if)#no switchport
R1(config-if)#ip address 10.0.0.1 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#interface FastEthernet2/4
R1(config-if)#no switchport
R1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/4, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/4, changed state to up

R1(config-if)#ip address 10.0.1.1 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#interface FastEthernet2/5
R1(config-if)#no switchport
R1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/5, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/5, changed state to up

R1(config-if)#ip address 10.0.2.1 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#interface FastEthernet2/6
R1(config-if)#no switchport
R1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/6, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/6, changed state to up

R1(config-if)#ip address 10.0.3.1 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#
```

54.

```
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 8 subnets, 2 masks
C       10.0.0.0/24 is directly connected, FastEthernet2/3
L       10.0.0.1/32 is directly connected, FastEthernet2/3
C       10.0.1.0/24 is directly connected, FastEthernet2/4
L       10.0.1.1/32 is directly connected, FastEthernet2/4
C       10.0.2.0/24 is directly connected, FastEthernet2/5
L       10.0.2.1/32 is directly connected, FastEthernet2/5
C       10.0.3.0/24 is directly connected, FastEthernet2/6
L       10.0.3.1/32 is directly connected, FastEthernet2/6
```

55.

```
C:\>ping 10.0.2.10

Pinging 10.0.2.10 with 32 bytes of data:

Request timed out.
Reply from 10.0.2.10: bytes=32 time<1ms TTL=127
Reply from 10.0.2.10: bytes=32 time<1ms TTL=127
Reply from 10.0.2.10: bytes=32 time=3ms TTL=127

Ping statistics for 10.0.2.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 3ms, Average = 1ms

C:\>tracert 10.0.2.10

Tracing route to 10.0.2.10 over a maximum of 30 hops:

  0  0 ms    0 ms    0 ms    10.0.1.1
  1  1 ms    0 ms    1 ms    10.0.2.10

Trace complete.
```

56:

```
C:\>ping 10.1.2.10

Pinging 10.1.2.10 with 32 bytes of data:

Reply from 10.0.1.1: Destination host unreachable.
Reply from 10.0.1.1: Destination host unreachable.
Reply from 10.0.1.1: Destination host unreachable.
Reply from 10.0.1.1: Destination host unreachable.

Ping statistics for 10.1.2.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>S|
```

57. R2:

```
R2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#interface FastEthernet2/4
R2(config-if)#no switchport
R2(config-if)#ip address 10.1.0.1 255.255.255.0
R2(config-if)#no shutdown
```

R3:

```
R3>ena
R3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#interface FastEthernet2/3
R3(config-if)#no switchport
R3(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/3, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/3, changed state to up

R3(config-if)#ip address 10.1.1.2 255.255.255.0
R3(config-if)#no shutdown
R3(config-if)#
```

R4:

```
R4>ena
R4#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R4(config)#interface FastEthernet2/4
R4(config-if)#no switchport
R4(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/4, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/4, changed state to up

R4(config-if)#ip address 10.1.2.1 255.255.255.0
R4(config-if)#no shutdown
R4(config-if)#interface FastEthernet2/5
R4(config-if)#no switchport
R4(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/5, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/5, changed state to up

R4(config-if)#ip address 203.0.113.1 255.255.255.0
R4(config-if)#no shutdown
R4(config-if)#interface FastEthernet2/6
R4(config-if)#no switchport
R4(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/6, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/6, changed state to up

R4(config-if)#ip address 10.1.3.1 255.255.255.0
R4(config-if)#no shutdown
R4(config-if)#
```

58:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.1.2.1

Pinging 10.1.2.1 with 32 bytes of data:

Reply from 10.1.2.1: bytes=32 time=1ms TTL=255
Reply from 10.1.2.1: bytes=32 time<1ms TTL=255
Reply from 10.1.2.1: bytes=32 time<1ms TTL=255
Reply from 10.1.2.1: bytes=32 time<1ms TTL=255

Ping statistics for 10.1.2.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>|
```

59: R1:

```
R1>ena
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#ip route 10.1.0.0 255.255.255.0 10.0.0.2
R1(config)#ip route 10.1.1.0 255.255.255.0 10.0.0.2
R1(config)#ip route 10.1.2.0 255.255.255.0 10.0.0.2
R1(config)#ip route 10.1.3.0 255.255.255.0 10.0.0.2
R1(config)#end
R1#

R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 12 subnets, 2 masks
C       10.0.0.0/24 is directly connected, FastEthernet2/3
L       10.0.0.1/32 is directly connected, FastEthernet2/3
C       10.0.1.0/24 is directly connected, FastEthernet2/4
L       10.0.1.1/32 is directly connected, FastEthernet2/4
C       10.0.2.0/24 is directly connected, FastEthernet2/5
L       10.0.2.1/32 is directly connected, FastEthernet2/5
C       10.0.3.0/24 is directly connected, FastEthernet2/6
L       10.0.3.1/32 is directly connected, FastEthernet2/6
S       10.1.0.0/24 [1/0] via 10.0.0.2
S       10.1.1.0/24 [1/0] via 10.0.0.2
S       10.1.2.0/24 [1/0] via 10.0.0.2
```

R2:

```
R2>ena
R2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#ip route 10.0.1.0 255.255.255.0 10.0.0.1
R2(config)#ip route 10.0.2.0 255.255.255.0 10.0.0.1
R2(config)#ip route 10.0.3.0 255.255.255.0 10.0.0.1
R2(config)#ip route 10.1.1.0 255.255.255.0 10.1.0.1
R2(config)#ip route 10.1.1.0 255.255.255.0 10.1.0.2
R2(config)#ip route 10.1.2.0 255.255.255.0 10.1.0.2
R2(config)#ip route 10.1.3.0 255.255.255.0 10.1.0.2
R2(config)#end
```

R3:

```
R3>ena
R3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#ip route 10.0.0.0 255.255.255.0 10.1.0.2
R3(config)#ip route 10.0.1.0 255.255.255.0 10.1.0.2
R3(config)#ip route 10.0.2.0 255.255.255.0 10.1.0.2
R3(config)#ip route 10.0.3.0 255.255.255.0 10.1.0.2

R3(config)#end
```

R4:

```
R4>ena
R4#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R4(config)#ip route 10.1.0.0 255.255.255.0 10.1.1.1
R4(config)#ip route 10.0.0.0 255.255.255.0 10.1.1.1
R4(config)#ip route 10.0.1.0 255.255.255.0 10.1.1.1
R4(config)#ip route 10.0.2.0 255.255.255.0 10.1.1.1
R4(config)#ip route 10.0.3.0 255.255.255.0 10.1.1.1
R4(config)#end
R4#
%SYS-5-CONFIG_I: Configured from console by console
```

60

61:

```
R1>ena
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#no ip route 10.1.0.0 255.255.255.0 10.0.0.2
R1(config)#no ip route 10.1.1.0 255.255.255.0 10.0.0.2
R1(config)#no ip route 10.1.2.0 255.255.255.0 10.0.0.2
R1(config)#no ip route 10.1.3.0 255.255.255.0 10.0.0.2
R1(config)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console
```

62:

```
%SYS-5-CONFIG_I: Configured from console by console

R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#ip route 10.1.0.0 255.255.255.0 10.0.0.2
R1(config)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console
```

63:

64:

```
R5>ena
R5#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R5(config)#interface FastEthernet2/3
R5(config-if)#no switchport
R5(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/3, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/3, changed state to up

R5(config-if)#ip address 10.1.3.2 255.255.255.0
R5(config-if)#no shutdown
R5(config-if)#interface FastEthernet2/4
R5(config-if)#no switchport
R5(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/4, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/4, changed state to up

R5(config-if)#ip address 10.0.3.2 255.255.255.0
```

65:

```
C:\>ping 10.0.3.2

Pinging 10.0.3.2 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.0.3.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

66.

```
R1>ena
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#ip route 10.1.3.0 255.255.255.0 10.0.3.2
R1(config)#end

R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 10 subnets, 2 masks
C       10.0.0.0/24 is directly connected, FastEthernet2/3
L       10.0.0.1/32 is directly connected, FastEthernet2/3
C       10.0.1.0/24 is directly connected, FastEthernet2/4
L       10.0.1.1/32 is directly connected, FastEthernet2/4
C       10.0.2.0/24 is directly connected, FastEthernet2/5
L       10.0.2.1/32 is directly connected, FastEthernet2/5
C       10.0.3.0/24 is directly connected, FastEthernet2/6
L       10.0.3.1/32 is directly connected, FastEthernet2/6
S       10.1.0.0/24 [1/0] via 10.0.0.2
S       10.1.3.0/24 [1/0] via 10.0.3.2

R1#
```

67:

```
C:\>ping 10.0.3.2

Pinging 10.0.3.2 with 32 bytes of data:

Reply from 10.0.3.2: bytes=32 time=32ms TTL=254
Reply from 10.0.3.2: bytes=32 time=9ms TTL=254
Reply from 10.0.3.2: bytes=32 time<1ms TTL=254
Reply from 10.0.3.2: bytes=32 time<1ms TTL=254

Ping statistics for 10.0.3.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 32ms, Average = 10ms

C:\>
```

68; R4:

```
R4(config-if)#interface FastEthernet2/5
R4(config-if)#no switchport
R4(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/5, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/5, changed state to up

R4(config-if)#ip address 203.0.113.1 255.255.255.0
R4(config-if)#no shutdown

R4#show ip interface brief

```

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/1	unassigned	YES	NVRAM	administratively down	down
FastEthernet2/3	unassigned	YES	unset	up	up
FastEthernet2/4	10.1.2.1	YES	manual	up	up
FastEthernet2/5	203.0.113.1	YES	manual	up	up
FastEthernet2/6	10.1.3.1	YES	manual	up	up
GigabitEthernet2/1	unassigned	YES	unset	down	down
GigabitEthernet2/2	unassigned	YES	unset	down	down
Dot11Radio2/1	unassigned	YES	NVRAM	administratively down	down
Vlan1	unassigned	YES	NVRAM	up	up

```
R4#
```

69:

```
R4#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R4(config)#ip route 0.0.0.0 0.0.0.0 203.0.113.2
R4(config)#end
R4#
%SYS-5-CONFIG_I: Configured from console by console
```

70:

71:

Yes. R1 automatically has routes for every network that is directly connected to its interfaces. These routes are added automatically when the interfaces are configured with IP addresses and are up. They appear as C (Connected) routes in the routing table.

72:

PC1 loses reachability to PC3 because R1 no longer has a route to PC3's network. PC1 and PC3 have not changed, but R1 needs a route to know where to forward the traffic. Without the static route, R1 cannot reach PC3.

73:

A summary route combines several smaller networks into one larger network. This makes the routing table smaller and easier to manage.